

A stronger test of the frozen DMDE scenarios

Version 0.9.20 - five-channel weak-rate closure and differential-response design

In More Basic Terms

This release improves the checks used to test a dark-matter and dark-energy hypothesis. It keeps two proposed particle-injection scenarios fixed and makes it harder for a computer calculation to pass while leaving out important reactions or failing to carry the injected particles through the calculation correctly. It also shows how closely the two scenarios are matched and estimates the numerical precision needed to compare their effects. These are improvements to the test and its reliability. They do not yet show that the hypothesis explains dark matter, dark energy, or a unified theory of everything. The next step is to run the complete particle-transport and early-universe element-formation calculations, validate the returned evidence, and compare independent implementations.

What changed

A reproduced omission now fails. The prior synthetic counterexample left out electron and positron capture and used vacuum beta decay at every temperature. The new checks test all five exported process histories.

The calculation must link its stages. The contract ties the injected source to the transported state, weak rates, and the element-formation calculation. Small diagnostic perturbations check that the declared inputs are consumed.

The comparison has a defined precision target. The two fixed scenarios differ mainly in neutrino-source amplitude. The source-only analysis quantifies their small electromagnetic difference and the accuracy needed to compare future outputs.

Scientific status

This is a validation and numerical-design release. No production helium abundance (Y_p), deuterium-to-hydrogen ratio (D/H), or effective neutrino number (N_{eff}) is supplied. Passing the checks does not establish the physical correctness of a backend or confirm the hypothesis.

What the release supports

The original provider archive is preserved byte for byte. This note adds context without retuning the source scenarios.

Five-channel acceptance checks

Independent, neutron-lifetime-normalized Born histories supply audit envelopes for the five exported processes. Non-negligible processes must lie within 25%; each direction total must lie within 10%. The grid must resolve the 1.80433131 MeV antineutrino-capture threshold and the 0.78233341 MeV beta-decay endpoint. These are execution checks, not a full corrected weak-rate calculation.

Frozen-pair comparison

Source-level quantity	Value
Neutrino-amplitude ratio, second / first	1.0086922376507683
Electromagnetic-amplitude ratio	1.0000039859789593
Log-contrast ratio, electromagnetic / neutrino	0.000460556843

A conditional planning example: a response elasticity of 0.01 would require about 20.40 parts per million independent Gaussian uncertainty at each endpoint for an approximate 3-sigma comparison. This is not achieved production precision, observed significance, or an observational error bar. The source geometry alone does not bound the electromagnetic response.

What remains to be established

Run both blinded scenarios through an instrumented momentum-resolved transport and BBN backend; verify coarse/fine refinement, source witnesses, rate closure, and network-consumption evidence; then review and replicate the results independently. The five-column export does not separately include the inverse three-body process $p + e^- + \bar{\nu}_e \rightarrow n$. Consistent evidence alone cannot establish honest execution or complete physical corrections.

Sources and provenance

Source: the accompanying provider archive, especially README_PROVIDER_DISPATCH.md and the documents ending WEAK5_CLOSURE_AND_FALSE_PASS.md, FROZEN_PAIR_DIFFERENTIAL_RESPONSE_RESULT.md, and VALIDATION_SCOPE_AND_LIMITS.md. Fresh local checks are recorded in DMDE_v0920_PUBLIC_VERIFICATION_2026-09-05.json.

Earlier record identified by the author: zenodo.org/records/20187862. This note does not assert that the next version has been published.